CS640A - Report
Monotone Circuits for Connectivity Require Super-Logarithmic Depth

Ananta Mukherjee, Chandra Shekhar Tiwari, Neelabjo Shubhashis Choudhury
170102, 20111264, 19211264
anantam@iitk.ac.in, chandrat20@iitk.ac.in, neelabjo@iitk.ac.in

December 2, 2020

Abstract

We say that a Boolean function $f : \{0,1\}^n \to \{0,1\}$ is a monotone function if for input strings $x, y \in \{0,1\}^n$, $x \preccurlyeq y \Leftrightarrow f(x) \leq f(y)$. Here $x \preccurlyeq y$ iff $x_i \leq y_i$, $\forall i \in [n]$. A Boolean circuit is said to be a monotone circuit iff it does not have any NOT gate in it. Now, monotone functions and monotone circuits are equivalent i.e. for every monotone function there is a corresponding monotone circuit and a monotone circuit represents a monotone function. Now, s-t connectivity in a graph is monotone since if one path is there no matter what one does with the other edges the output is always 1. In this paper, authored by *Mauricio Karchmer* and *Avi Wigderson*, published in *SIAM J. Discrete Math., 3(2), 255–265*, the authors proved that every monotone circuit that decides $st - connectivity$ of an undirected graph over n vertices has depth $\Omega(\log^2 n)$. This implies that every monotone formula for the $st - connectivity$ of an n vertex graph has $n^{\Omega(\log n)}$ size.

1 Introduction

In this paper [**1**], the authors have shown an equivalence between circuit depth and the communication complexity of a certain related problem. Exploiting this relation, they have designed a strong communication game and have established a lower bound for the complexity of that game. This eventually gives a lower bound on the depth of the monotone circuit. The paper is organized into **three** parts:

1. **Define a communication game and its equivalence to circuit depth.**
2. **Proof of a depth analogue of Khrapchenko's theorem.**
3. **Lower bound for *st-connectivity*.**

2 Major Concepts and Tools

2.1 Communication Complexity and Circuit Depth

Circuits are considered over the basis $\{\wedge, \vee, \neg\}$ with following constraints:

- All $\{\wedge, \vee\}$-gates have fan-in 2.
- $\neg$-gates are only applied to input variables.

For a function f, let $d(f)$ denote the minimum depth of a circuit computing f.

2.1.1 Communication Game

There are two players P_1 and P_2. Let $B_0, B_1 \subseteq \{0,1\}^n$, such that $B_0 \cap B_1 = \phi$. P_1 gets $x \in B_1$, and P_2 gets $y \in B_0$. Now their **goal** is to find a coordinate i such that $x_i \neq y_i$. Let $C(B_1, B_0)$ be the minimum number of bits they have to communicate in order for both of them to agree on such a coordinate.

Theorem 2.1

For every function $f : \{0,1\}^n \to \{0,1\}$, we have; $d(f) = C(f^{-1}(1), f^{-1}(0))$

The proof of the above theorem follows from the following two lemmas.

Lemma 2.1

For all functions f and all $B_0, B_1 \subseteq \{0,1\}^n$, such that $B_0 \subseteq f^{-1}(0)$ and $B_1 \subseteq f^{-1}(1)$, we have; $C(B_1, B_0) \leq d(f)$

Proof: By induction on $d(f)$.

If $d(f) = 0$ then f is either x_i or $\overline{x_i}$. In either case, we have that for all $x \in B_1$ and $y \in B_0$, $x_i \neq y_i$, so that i is always an answer and $C(B_1, B_0) = 0$

For the induction step we suppose that $f = f_1 \wedge f_2$ (the case $f = f_1 \vee f_2$ is treated similarly) so that $d(f) = \max[d(f_1), d(f_2)] + 1$. Let $B_0^j = B_0 \cap f_j^{-1}(0)$ for $j = 1, 2$. By induction we have that $C(B_1, B_0^j) \leq d(f_j)$ for $j = 1, 2$. Consider the following protocol for B_1 and B_0: P_2 sends a 0 if $y \in B_0^1$, otherwise he sends a 1; the players then follow the best protocol for each of the sub-cases. We have

$$C(B_1, B_0) \leq 1 + \max_{j=1,2}[C(B_1, B_0^j)] \leq 1 + \max_{j=1,2}[d(f_j)] = d(f).$$

Lemma 2.2

let $B_0, B_1 \subseteq \{0,1\}^n$ such that $B_0 \cap B_1 = \phi$. Then, there exists a function f with $B_0 \subseteq f^{-1}(0)$ and $B_1 \subseteq f^{-1}(1)$ such that; $d(f) \leq C(B_1, B_0)$

Proof: By induction on $C(B_1, B_0)$. The details we skip here in this report. The technique is exactly the opposite to that of Lemma 2.1 i.e. here break our set B_0 into 2 disjoint subsets and based on which subset we go we build our function inductively.

2.1.2 Communication Game (Monotone Version)

There are two players P_1 and P_2. Let $P, Q \subseteq 2^{[n]}$ be such that $\forall p \in P \, \forall q \in Q, p \cap q \neq \phi$. P_1 gets $p \in P$, and P_2 gets $q \in Q$. And, their **goal** is to find an element in $p \cap q$. Let $C_m(P, Q)$ be the minimum number of bits they have to communicate in order to find such an element.

A modified version of theorem 2.1 can be used to capture the restrictions of monotone computation. A *minterm (maxterm)* of a monotone function f is a minimal set of variables which if we set to 1(0), f will be equal to 1(0) regardless of the other variables. Let $min(f)$, and $Max(f)$ denote the set of *minterms* and *maxterms* of a function f respectively. It is easy to see that every *minterm* intersects every *maxterm*. Also, let $d_m(f)$ denotes the minimum depth of a monotone circuit computing f.

Theorem 2.2

For every monotone functions f we have; $d_m(f) = C_m(min(f), Max(f))$.

Proof: If we apply the base case of lemma 2.1 on monotone circuits, we always find a coordinate i such that $x_i = 1$ while $y_i = 0$. On the other hand, if the protocol always gives a coordinate i with the above property, lemma 2.2 gives a monotone circuit.

let $x \in f^{-1}(1)$ be the characteristic vector of a subset $p \subseteq [n]$. Similarly, let $y \in f^{-1}(0)$ be the characteristic vector of the complement of a subset $q \subseteq [n]$ (i.e. $i \in q \iff y_i = 0$). By the above argument, it is clear that the answer of the protocol will be an element of $p \cap q$.

2.1.3 Stronger Communication Game

Let us assume P_1 is given $f^{-1}(0)$ and P_2 is given $f^1(1)$. Hence, to find an element of $p \cap q$, P_1 and P_2 can communicate amongst themselves by using a single bit to agree on that element. A similar communication game can be created to make both players powerful by making them send more than 1 bit in a single round. Specifically P_1 can send 2^a bits to P_2 and P_2 can be allowed to send a bits back.

Theorem 2.3

For every function f, there exists a protocol D where at each round player P_2 sends 2^a bits while player P_1 responds with a bits and such that the number k of rounds satisfies $k \leq \frac{d(f)}{a}$ (for the general game), and $k \leq \frac{d_m(f)}{a}$ (for the monotone version)

Proof: Let C be the best circuit for f. The idea is to simulate 'a' layers of C with a round of D. Divide C into stages of depth 'a' each and look at the sub circuit of each stage. Each one computes a function which depends on at most 2^a wires and, thus, can be represented in CNF form with at most 2^{2^a} clauses, each of length at most 2^a. Following the proof of lemma 2.1, it is easy to see that such a CNF representation can be simulated by a round where player P_2 sends 2^a bits and player P_1 send 'a' bits. The same holds for monotone case.

3 Khrapchenko's Bound

As a nice application of a theorem 2.1, we give a simple proof of a depth analogue of a theorem of Khrapchenko. Let C_n be the graph of the $n-$cube with vertex set $\{0, 1\}^n$ and those nodes

adjacent if and only if they differ in one coordinate. Any subset A of edges induces a graph G_A of C_n in the natural way. For a graph G_A and a node x, we denote $d_A(x), N_A(x)$ as the degree of x in A, and the set of neighbors of x in A respectively. We drop the sub-index A if no confusion arises. Let E denote expectation with uniform distribution.

Theorem 3.1 (Khrapchenko)

Let $B_0, B_1 \subseteq \{0,1\}^n$ such that $B_0 \cap B_1 = \phi$. Let $A = C_n \cap (B_0 \times B_1)$. Then for every function f with $B_0 \subseteq f^{-1}(0)$ and $B_1 \subseteq f^{-1}(1)$ we have
$d(f) \geq \log \frac{|A|^2}{|B_0||B_1|}$

Proof: Fix a protocol D for the communication game and let $C(x,y)$ be the number of bits D uses on input x, y. We will prove that for (x,y) taken uniformly from A we have

$$E(C(x,y)) \geq \log \frac{|A|^2}{|B_0||B_1|} \tag{$*$}$$

so that by lemma 2.1, we get $d(f) \geq E(C(x,y))$.
We view $(*)$ as follows: Write

$$\log \frac{|A|^2}{|B_0||B_1|} = \log \frac{|A|}{|B_0|} + \log \frac{|A|}{|B_1|}$$

and notice that $\frac{|A|}{|B_0|}$ and $\frac{|A|}{|B_1|}$ are the average degrees of nodes in B_0 and B_1 respectively. In what follows, we will claim that the number of bits P_1 sends is at least the logarithm of the average degree of nodes in B_1 (similarly with P_2). Intuitively, this is so because even if P_1 knows y, he needs $\log d(y)$ bits to tell P_2 which x he has.

We now proceed formally. For $(x,y) \in A$, let $b_1(x,y)$, $b_2(x,y)$ be the number of bits P_1 and P_2 send when the input of the protocol is (x,y). We have $E(C(x,y)) = \frac{1}{|A|}\left[\sum_{(x,y)\in A}(b_1(x,y) + b_2(x,y))\right] = \frac{1}{|A|}\left[\sum_{x\in B_1}\sum_{y\in N(x)}(b_2(x,y) + \sum_{y\in B_0}\sum_{x\in N(y)}(b_1(x,y))\right]$. Now, we claim: For any $x \in B_1$, $\sum_{y\in N(x)} b_2(x,y) \geq d(x)\log d(x)$. This is so because, even if P_2 knows x, he has to tell P_1 which y he has. Similarly, $\forall y \in B_0$ we have $\sum_{x\in N(y)} b_1(x,y) \geq d(y)\log d(y)$.

We now conclude, $E(C(x,y)) \geq \frac{1}{|A|}\left(\sum_{x\in B_1} d(x)\log d(x) + \sum_{y\in B_0} d(y)\log d(y)\right)$
$\geq \frac{1}{|A|}\left(\sum_{x\in B_1}\frac{|A|}{|B_1|}\log\frac{|A|}{|B_1|} + \sum_{y\in B_0}\frac{|A|}{|B_0|}\log\frac{|A|}{|B_0|}\right) = \log\frac{|A|^2}{|B_1||B_0|}$ where the last inequality follows from convexity of $x\log x$.

4 Intuition of s-t connectivity and depth of a monotone circuit

Taking a layers of the circuit in one round and then finding the child on which both players agree and continue till a variable in the leaf node is reached. Given Theorem 2.3 we can define the communication game between P_1 and P_2 where they try to agree on a specific edge in a graph. P_1 has an *s-t path* p which is a minterm and P_2 has a coloring $q : V \to \{0, 1\}$ for all nodes in the graph (Start vertex *q(s)* is colored 0 and *q(t)* is colored 1). Naturally, a **bichromatic edge** is an edge from a color 0 to a color 1 vertex. This edge would definitely belong to an *s-t cut* which is a maxterm. This intuition leads us to find such an edge which lies both in *s-t path* and is bichromatic and happens when both players agree after a round. Only one such edge needs to be found and thus, P_1 and P_2 need to agree just once.

What follows trivially is a divide and conquer strategy that P_1 follows. This player can send the middle vertex in its *s-t path*, which takes $log(n)$ bits. P_2 responds with a single bit which denotes color of this vertex. We can understand that even if P_2 replies with more information, say n^ϵ bits, it may not help reduce the number of rounds. This is because P_2 may be sending irrelevant information about vertices which it presumes to be in the *s-t path* of P_1. Since our objective is to find the minimum number of rounds for agreement between the two, P_1 should ask relevant questions and P_2 should answer more for such questions.

Hence, a new game *stconn(l)* is defined, where P_1's *s-t path* has length l. In this setting, P_1 can send $\epsilon log(n)$ many bits and P_2 replies with n^ϵ bits. This sets the stage to construct the main theorem of the paper which gives us the minimum number of rounds required for agreement, thereby establishing a lower bound on the depth of the monotone circuit for s-t connectivity.

Theorem 4.1

Suppose $l \leq n^{\frac{1}{10}}$. There exists an ϵ with $0 < \epsilon < \frac{1}{2}$ such that if D is a k-round protocol for stconn(l) where at each round player I sends $\epsilon \log n$ bits and player II sends n^ϵ bits, then then $k \geq \log l$.

Corollary 4.1

Any monotone circuit computing st-connectivity requires $\Omega(\log^2(n))$ depth.

Proof : Since $k \geq \log l$, so $k \geq \delta \log(n)$. From Theorem 2.3, on substituting a with $\epsilon \log(n)$, $d_m(f) \geq a.k = (\epsilon \log(n)) \cdot (\delta \log(n))$, i.e, $d_m(f) \geq \epsilon\delta \log^2(n)$. Thus depth of the monotone circuit is super-logarithmic, i.e, has a tight bound of $\Omega(\log^2(n))$. The vital question is why Theorem 4.1? It follows from the following stronger theorem, namely 4.2. We discuss some intuitions and strategies and follow it with it's proof.

Corollary 4.2

Any monotone formula for st-connectivity has size $n^{\Omega(\log n)}$.

Theorem 4.2

Suppose $l \leq n^{\frac{1}{10}}$. There exists an ϵ with $0 < \epsilon < \frac{1}{2}$ s.t. if D is a k round protocol where in each round player I sends $\epsilon \log n$ bits and player II responds with n^{ϵ} bits, and D solves the game vector(l) for at least a fraction of $\frac{1}{8}n^{-\epsilon}$ of all possible vectors, then $k \geq \log l$.

This is a stronger version of Theorem 4.1. which states that if the protocol D solves the game for all possible l vectors, then lower bound on number of rounds is true. Whereas, Theorem 4.2 states that instead of all l vectors if D solves it for a large enough subset of all l vectors, the same lower bound would still hold true.

5 The Final Proof

To prove Theorem 4.2, the authors use method of contradiction. Let us assume less than $\log l$ many rounds would suffice for the game. The first round is analyzed to reduce the problem size. We can argue that if the reduced problem can be solved in $k-1$ rounds with **sufficiently largeness** properties of the subsets involved, then **something impossible** can also be solved. This sets up the contradiction and will be the strategy.

Setup and Some Parameters : Since, we are assuming that less than $\log l$ many round is sufficient, we set $t_{max} = \log l - 1$, t actually counts the no of rounds. Let, n_t denotes the size of universe (a subset of all vertices) at the tth round and l_t denotes the length of paths that we care about at the tth round.

Let, $n_0 = n$ and $n_{t+1} = n_t - 4n_t^{\frac{1}{2}}$. We see, $\frac{n}{2} \leq n_t \leq n$ for $t \leq t_{max}$ AND $l_0 = l$ and $l_{t+1} = \frac{l_t}{2}$. We see, $2 \leq l_t \leq l$ for $t \leq t_{max}$. We consider the following proposition:

H(t,k): There exists a collection of vectors $P^t \subseteq [n_t]^{l_t}$ of length l_t and a collection of colorings $Q_t \subseteq \{0,1\}^{n_t}$ with $\mu(P^t) \geq \frac{1}{8}n^{-\epsilon}$ and $\mu(Q_t) \geq 2^{-2tn^{\epsilon}}$ such that there exists a k-round protocol D^t good for (P^t, Q^t). We have the following vital claims :

Claim 4.1 For $t \leq t_{max}$, $\neg H(t, 0)$.

Claim 4.2 For $t \leq t_{max}$, $H(t,k) \rightarrow H(t+1, k-1)$.

Before going into the proof details, let us first see why they help. They simultaneously imply $\neg H(0, t_{max})$. This is Theorem 4.2 itself because $\neg H(0, t_{max})$ tells that at level 0 we cannot have a $t_{max} = \log l - 1$ round protocol i.e. $\log l$ is a lower bound on no of rounds. But why this implication? Because if $H(0, t_{max})$ then by claim 4.2 $H(1, t_{max}-1)$, $H(2, t_{max}-2)$,... and $H(t_{max}, 0)$ which directly contradicts claim 4.1.

Claim 4.1 is true because at any stage t, with sufficiently large subsets of paths and colors we cannot have a 0 round protocol. Because 0 round protocol means I and II knows each others' information which we don't allow. Claim 4.2 is not trivial and requires a long and involved proof. Specifically, at this stage, it is not very clear why with one transition, along with reducing the no of round by 1, we must find the required subsets with large enough densities.

First we take a brief detour to discuss a very crucial combinatorial result due to Hastad in [**5**] and several of its relevant corollaries. We skip the proofs here. Here whenever we talk about density (μ) of a subset we mean the ratio of its size to that of the universe. Here is the setting. Let, $H \subseteq A_1 \times ... \times A_k$ and for $v \in A_i$, $Ext_{A_i}(v) = \{u \in H : u_i = v\}$. Note that, even if $Ext_{A_i}(v) \subseteq H$,

its universe is $H_i/A_i = A_1 \times ... \times A_{i-1} \times A_{i+1} \times ... \times A_k$.

Lemma 4.1, Hastad's Lemma: Let, $H \subseteq A_1 \times ... \times A_k$. Let, $B_i = \{u \in A_i : \mu(Ext_{A_i}(v)) \geq \mu(H)/2k\}$. Then $\prod_{i=1}^{k} \mu(B_i) \geq \frac{\mu(H)}{2}$.

Corollary 4.3: If k=2, then there exists an i such that $\mu(B_i) \geq (\frac{\mu(H)}{2})^{1/2}$.

Corollary 4.4: If an i is chosen randomly from [k], then $Pr[\mu(B_i) < (\mu(H)/2)^{2/k}] < 1/2$.

Now, let's return to the proof of Claim 4.2. Recall, assuming instances of P^t, Q^t, D^t we are required to give instances of $P^{t+1}, Q^{t+1}, D^{t+1}$. Here Pigeon Hole Principle helps. Since, I can send $\epsilon \log n$ many bits in first round, at most $2^{\epsilon \log n} = n^{\epsilon}$ many variations can be created. Thus, there exists some $P \subseteq P^t$ with $\mu(P) \geq \mu(P^t).n^{-\epsilon} \geq \frac{1}{8}n^{-2\epsilon}$ s.t. for every path p $\in$ P, player I sends the same bits in round 1. Similarly, there exists a coloring $Q \subseteq Q^t$, with $\mu(Q) \geq 2^{-(2t+1)n^{\epsilon}}$ s.t. for all coloring $q \in Q$, player II sends the same bits in round I.

Now, let $L = \{1, ..., l_t/2\}$ and $R = \{l_t/2+1, ..., l_t\}$. We say that P is L-good if many left projections of P, each have many extensions to the right, formally if $\mu(\{p_L : \mu(Ext_{P,L}(p_L)) \geq n^{-2\epsilon}/32\}) \geq \frac{1}{4}n^{-\epsilon}$. Why this strange definition? Because L length paths are nothing but subset of Cartesian product of their left and right halves. So, in the setting of Hastad lemma if we set $\mu(H) = n^{-2\epsilon}/8$, the density lower bound of P, this follows. The R-goodness can be defined similarly.

Now, the crucial thing is, using the corollary 4.3 tells us that P is either L-good or R-good. W.l.o.g, let P be L-good and A be the set of vertices in P_L with many extensions. The concept of restriction of coloring is introduced thereafter. Restriction fixes a color, either 0 or 1, for a vertex and this leads us to our final lemma.

Lemma 4.3: There exists a restriction $\rho : [n_t] \to \{0, 1, *\}$ with $\rho^{-1}(*) = n_{t+1}$ s.t. the following properties hold good:
G1: $\mu(Q^{\rho}_{\rho^{-1}(*)}) \geq 2^{-2(t+1)n^{\epsilon}}$ i.e. out of all colorings in Q, if we take only those that obey the restriction ρ and project them to the free vertices, we get a subset of coloring for the free vertices with sufficiently large density.
G2: $\exists \overline{P} \subseteq P$ s.t.

- $\forall p \in \overline{P}$, $p_L \subseteq \rho^{-1}(*)$ and $p_R \subseteq \rho^{-1}(1)$.
- $\forall p, p' \in \overline{P}, p_L \neq p'_L$
- $\mu(\overline{P}_L) \geq \frac{1}{8}n^{-\epsilon}$

Now, this gives that in $\overline{P}$, all bichromatic edges must lie in the LHS since all vertices RHS are pre-colored with 1. Thus, we set $P^{t+1} = \overline{P}_L$, $Q^{t+1} = Q^{\rho}_{\rho^{-1}(*)}$ and D^{t+1} on (P^{t+1}, Q^{t+1}) simulates the remaining rounds of D^t on $(\overline{P}, Q^{\rho})$.

Finally, the question is why lemma 4.3 holds? This is indeed long and convoluted. To be very brief we assume the contrary first i.e. we assume there is no such restriction. Then using probabilistic methods we show that $Pr(\neg G1) + Pr(\neg G2) < 1$ i.e. there is indeed some restriction that satisfies G1 and G2.

6 Conclusion, Related works and an Open problem

While proving Theorem 2.3, we used the idea of creating a communication game for finding agreement between two players. This helps us understand that such an approach to calculate lower bounds, fundamentally differ from those of finding upper bounds. We basically need to calculate minimum values of all factors affecting depth of an unique monotone circuit, defined according to the problem we need to solve. And a strong communication game between 2 players help us achieve exactly this.

This notion of communication game can be used on graph G with maximum bipartite clique size K and the goal is to identify a non-edge between A and B. Motivated in [**3**], the authors consider a communication game where two parties separately receive disjoint subsets A, B of vertices such that $|A| + |B| = K$. The eventually prove that O(log n) bits of communication are enough for any n-vertex graph to solve this problem. A lot of work has been done to obtain lower bounds for complexity of monotone functions. In [**4**], the authors give a general bound of the monotone complexity of a Boolean function by some of its combinatorial characteristics. They give a sequence of functions for which the obtained estimate has growth $2^{n^{(\frac{1}{3}-o(1))}}$. Like the separation of monotone formulas from circuits, authors in [**6**] separate bounded-depth formulas versus circuits and they use a lower bound for parameterized version of st-connectivity for that.

P_1 and P_2, in this paper, uses different properties of the edge they need to agree on, i.e, an **agreement edge**,. Specifically, P_1 uses the property that this **agreement edge** lies on an *st path* which it has with him and P_2 uses the property that it must belong to an *st cut* and she has the coloring for that. Proving a lower bound for *general connectivity* in a graph has similar properties and a similar approach may help establish the lower bound, but this is still an open problem.

References

[1] Mauricio Karchmer and Avi Wigderson, Monotone Circuits for connectivity require super-logarithmic depth, 1990 Society for Industrial and Applied Mathematics, SIAM J. DISC. MATH. Vol. 3, No. 2, pp. 255-265, May 1990.

[2] Benjamin Rossman, Formulas versus Circuits for Small Distance Connectivity. SIAM Journal on Computing 47:5, 1986-2028, 2018. Available online : https://doi.org/10.1137/15M1027310.

[3] Stasys Jukna, Clique problem, cutting plane proofs and communication complexity. Information Processing Letters 112:20, 772-777, October, 2012. Available online : https://doi.org/10.1016/j.ipl.2012.07.003.

[4] Andreev, A.E. A method for obtaining efficient lower bounds for monotone complexity. Algebra and Logic 26, 1–18 (1987). Available online : https://doi.org/10.1007/BF01978380.

[5] J. Håstad, Almost optimal lower bounds for small depth circuits. STOC '86 1986. Available online : https://doi.org/10.1145/12130.12132.

[6] Benjamin Rossman, Formulas versus Circuits for Small Distance Connectivity. SIAM J. Comput., 47(5), 1986–2028. Available online : https://doi.org/10.1137/15M1027310.